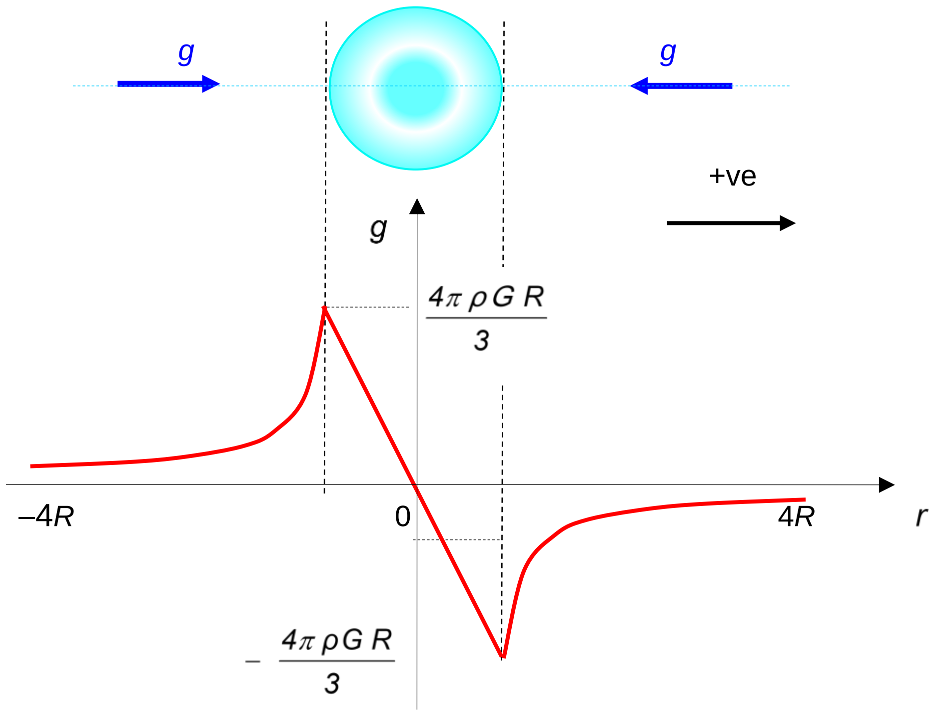
4	
(a)	$g = \frac{F}{m}$ *where* $F = \frac{GMm}{r^2}$ $= \frac{GM}{r^2}$
	as $\rho = \frac{M}{\frac{4}{3}\pi R^3}$
	$g = \frac{4\pi\rho\, G\, R^3}{3\, r^2}$
(b)	
	$-R < r < R$: straight line passing through the origin

	$r < -R$ and $r > R$: inverse square graph
	+ve and -ve direction of g (according to stated sign convention)
	$r = R: \quad g = \dfrac{4\pi \rho G R}{3}$
(c)(i)	By N2L, $m a = m g$ $a = \dfrac{4\pi \rho G}{3} r \quad \cdots\cdots(1)$ For SHM, $a = \omega^2 x \quad \cdots\cdots(2)$ Comparing $\omega^2 = \dfrac{4\pi \rho G}{3}$
	$\left(\dfrac{2\pi}{T}\right)^2 = \dfrac{4\pi \rho G}{3}$ $T = \sqrt{\dfrac{3\pi}{\rho G}}$
(c)(ii)	Maximum time taken (as assuming released from rest in Singapore) = half a period
	$\dfrac{T}{2} = \dfrac{1}{2}\sqrt{\dfrac{3\pi}{\rho G}}$ $= \dfrac{1}{2}\sqrt{\dfrac{3\pi}{\left(5.51\times10^{3}\right)6.67\times10^{-11}}}$ $= 2530\ \text{s}\ (3\ \text{s.f.})$
5	